

Exploring the Intersection Between STIX Framework and TARA for Sharing Cyber Threat Intelligence for Automotive Systems

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Abstract—The automotive sector is experiencing profound technological evolution with increasing digitalization, self-driving capability, and connectivity. These advancements, however, introduce critical cybersecurity challenges in modern vehicles. While standards such as ISO/SAE 21434:2021 and UN Regulation No. 155 mandate Threat Analysis and Risk Assessment (TARA), there remains a lack of standardized approaches for understanding and communicating automotive attacks, vulnerabilities, and security mechanisms. Cyber Threat Intelligence (CTI), although widely utilized in traditional IT domains, has not been extensively adopted in automotive contexts. This article investigates the convergence of STIX objects with TARA methodologies, examining their potential integration to enhance vehicular cybersecurity. Through systematic mapping between these frameworks, we analyze benefits, challenges, and implementation considerations. Our research demonstrates how STIX can strengthen threat characterization, vulnerability assessment, and mitigation planning while facilitating information sharing across automotive supply chains, addressing the need for standardized approaches to automotive cybersecurity intelligence.

Index Terms—Automotive cybersecurity, threat analysis, cyber threat intelligence, STIX framework, risk assessment, connected vehicles.

I. INTRODUCTION

The automotive industry is undergoing a rapid transformation, driven by trends towards digitalization, autonomous driving, connectivity, electrification, and shared mobility [1]. This transformation, while offering exciting possibilities, also introduces a significant challenge: cybersecurity. Modern vehicles are becoming increasingly complex cyber-physical systems, incorporating numerous Electronic Control Units (ECUs), intricate networks, and vast amounts of data [2]. Cybersecurity vulnerabilities in vehicles can lead to severe consequences that extend beyond data breaches to safety-critical impacts. Attacks targeting critical functions like steering, braking, or engine control can lead to catastrophic accidents. Understanding this significance, the automotive sector has embraced standards such as ISO/SAE 21434 [3] and the UR R155 regulation [4], which establish frameworks for addressing cybersecurity risks throughout the vehicle lifecycle. It offers a framework for defining cybersecurity process requirements throughout

the entire vehicle lifecycle. Clause 15 of ISO/SAE 21434 delineates the Threat Analysis and Risk Assessment (TARA) process for early threat identification during design phases.

In the IT domain, Cyber Threat Intelligence (CTI) has enabled organizations to adopt proactive security postures [5]. Structured Threat Information eXpression (STIX) is a language and serialization format used to exchange cyber threat intelligence (CTI) [6]. Created with an emphasis on expressiveness, adaptability, and machine readability, STIX enables the efficient sharing of threat data while remaining accessible to human analysts. By providing standardized formatting for CTI, STIX facilitates both threat analysis and collaborative information sharing across organizations. Despite these capabilities, significant adaptation challenges emerge when applying CTI to automotive systems due to their unique complexity. The automotive domain, with its intricate network of interconnected ECUs and specialized communication protocols, presents distinct obstacles for cybersecurity frameworks originally developed for traditional IT environments.

This research investigates the application of STIX to automotive TARA processes examining their intersection within the ISO/SAE 21434 framework. We focus on the benefits, challenges, and implementation considerations for enhancing automotive cybersecurity through structured threat intelligence.

The remainder of this paper is organized as follows: Section II examines existing research on automotive cybersecurity taxonomies; Section III provides an overview of CTI concepts and STIX; Section IV analyzes automotive network characteristics; Section V explores mapping STIX objects to TARA; and Sections VI and VII present discussion and conclusions.

II. RELATED WORK

Several research efforts have focused on developing comprehensive taxonomies to systematize the understanding of automotive cybersecurity. Thing and Wu [7] proposed a multi-dimensional taxonomy that categorizes automotive attacks based on attacker characteristics, attack vectors, targeted components, motivations, and potential consequences.

Pekaric et al. [8] extended this work by conducting a systematic literature review to classify attack techniques specifically targeting automotive systems. Their research mapped various attack methodologies to the Common Attack Pattern Enumeration and Classification (CAPEC) framework [9], particularly CAPEC-1000: Mechanisms of Attack.

Dürrwang [10] investigated the integration of safety and security analysis in automotive systems, proposing a Safety and Security Goal Model (SGM) that facilitates comprehensive risk assessment. The research also developed a detailed taxonomy of cyberattacks targeting vehicles, with particular attention to attacks that could impact safety-critical functions.

Sommer et al. [11] created a model-based security testing method (AMBST) alongside an Automotive Attack Database (AAD) that documents and categorizes real-world attacks on vehicles.

While substantial research exists on automotive cybersecurity taxonomies, there remains a significant gap in connecting standardized CTI frameworks like STIX with automotive-specific threat analysis processes such as TARA. The integration of these domains presents both opportunities and challenges that have not been thoroughly explored in existing literature. Our research addresses this gap by investigating the intersection between STIX and TARA within the context of ISO/SAE 21434, examining how standardized threat intelligence representation can enhance automotive cybersecurity practices.

III. CYBER THREAT INTELLIGENCE

CTI is actionable, evidence-based knowledge that provides context for understanding and mitigating cyberattacks [5]. This intelligence encompasses information regarding threat actors, their motivations, capabilities, methodologies, and indicators of compromise. CTI acquisition integrates diverse information sources including Open-Source Intelligence (OSINT), technical forums, and internal security data [12], which security practitioners analyze to inform strategic security decisions. Threat intelligence is classified into four primary categories based on scope, granularity, and intended audience [13].

Common categories include:

- **Strategic threat intelligence:** High-level insights into threat landscapes and trends for long-term security planning.
- **Operational threat intelligence:** Information about specific attack campaigns and adversary activities for imminent threat preparation.
- **Tactical threat intelligence:** Details of adversary TTPs for developing targeted defense mechanisms.
- **Technical threat intelligence:** Low-level indicators of compromise (IOCs) for automated detection and response.

A. Structured Threat Information Expression

STIX is a standardized language for representing and exchanging CTI in a machine-readable format [6], [14], facilitating collaborative analysis across organizational boundaries

[12]. Originally developed by MITRE Corporation and now maintained by OASIS, STIX has evolved through several iterations:

- **STIX 1.0:** Introduced in 2012, STIX 1 utilized an XML-based format with complex nested structures [15].
- **STIX 2.0:** Standardized in 2017, marked a paradigm shift by adopting a JSON-based format with a flatter, more relational data model.
- **STIX 2.1:** released in 2021, Current standard with expanded object types and relationships [6].

STIX supports essential cybersecurity functions including threat analysis, indicator pattern specification, and coordinated incident response management [15].

B. Structure of STIX

STIX implements a graph-based data model where threat intelligence is represented as a network of interconnected nodes and edges. STIX Domain Objects (SDOs) and STIX Cyber-observable Objects (SCOs) function as nodes within this graph structure. SDOs represent high-level intelligence concepts like attack patterns, campaigns, courses of action, and threat actors. SCOs, on the other hand, characterize observable facts about a network or host, such as file hashes, IP addresses, and URLs. The STIX architecture defines several key concepts as independent and reusable constructs [6], [15]:

- **Attack Pattern:** Provides a formal representation of common attack methodologies or techniques employed by threat actors. Each Attack Pattern object includes details about the attack execution steps, goals, and potential impact. These objects represent Tactics, Techniques, and Procedures (TTPs) and can be linked to external knowledge bases such as CAPEC [9].
- **Campaigns:** Represents coordinated series of malicious activities or attacks pursuing specific objectives over time. Campaigns are characterized by their objectives, the incidents they cause, the people or resources they target, and the resources they utilize.
- **Courses of Action (COAs):** Defines mitigation or response strategies designed to address specific threats, minimize their effects, and enhance security posture.
- **Indicators:** Encapsulates patterns of observable and contextual information intended to detect malicious activity. Indicators establish connections between observable patterns and the threats they represent.
- **Incidents:** Documents discrete occurrences of cyberattacks or security events affecting an organization. Incident objects incorporate data from indicators, including TTPs, affected assets, and comprehensive impact assessments.
- **Infrastructure:** Represents physical or virtual systems supporting cyber operations, whether offensive or defensive. This object can represent the command and control infrastructure, the malware-hosting servers, or the communication channels used by threat actors.
- **Vulnerability:** Identifies weaknesses or exposures in systems, applications, or networks that could be exploited

by attackers. Vulnerability objects typically reference industry-standard databases such as CVE and CWE [16], [17].

- **Tool:** Represents software applications or utilities employed in cyber operations. This category encompasses penetration testing frameworks, malware development kits, and other software tools leveraged by threat actors.
- **Malware:** Documents specific malicious software instances, including viruses, trojans, worms, ransomware, and other malicious code.
- **Threat Actors:** Characterizes individuals, groups, or organizations believed to operate with malicious intent. These objects capture comprehensive profiles including motivation, sophistication level, resource availability, and observed TTPs.

STIX Relationship Objects (SROs) define the semantic connections between nodes. The framework defines two primary SRO types:

- **Generic Relationship:** This SRO utilizes a *relationship_type* property to define the specific nature of connections between objects. The STIX specification provides a predefined vocabulary of relationship types, such as an indicator *indicates* a campaign. The framework also supports custom relationship types to accommodate specialized intelligence modeling requirements.
- **Sighting:** This specialized SRO documents instances where entities have observed specific SDOs, providing properties like count for quantitative context. Sightings establish temporal and spatial dimensions for threat intelligence.

STIX objects incorporate common properties that provide core capabilities for managing and contextualizing the information they represent. These properties apply consistently across STIX object types, including SDOs, SROs, and SCOs.

Table I summarizes the primary STIX objects relevant to our research, detailing their key properties and relationship types.

SROs define the type of relationship between SDOs and SCOs. The two primary SROs are:

- **Generic Relationship:** This SRO employs a *relationship_type* property to define the specific nature of the connection. The STIX specification provides a predefined set of relationship types, such as an indicator *indicates* a campaign. Users also have the flexibility to define their own custom relationship types to capture connections not covered in the standard.
- **Sighting:** This specialized SRO records instances where an entity has observed an SDO, offering properties like count to add further context.

STIX objects have a set of common properties that provide core capabilities for managing and understanding the information they represent. These properties apply to various STIX object types, including SDOs, SROs, and SCOs. The main STIX objects that we will use in our study are summarized

TABLE I: summary of STIX objects.

Object Type	Category	Key Properties	Relationships
Attack Pattern	SDO	name, description, aliases, kill_chain_phases	delivers, targets, uses
Course of Action (COA)	SDO	name, description	investigates, mitigates, remediates
Infrastructure	SDO	name, description, infrastructure_types, kill_chain_phases	communicates-with, consists-of, controls, delivers, has vulnerability, hosts
Malware	SDO	name, description, malware_types, is_family, aliases, kill_chain_phases, first_seen, last_seen	authored-by, communicates-with, compromises, exploits
Threat Actor	SDO	name, description, threat_actor_types, aliases, sophistication, resource_level, primary_motivation, secondary_motivations	attributed-to, hosts, owns, impersonates, targets, uses
Tool	SDO	name, description, tool_types, aliases, kill_chain_phases, tool_version	delivers, has vulnerability, targets, uses
Vulnerability	SDO	name, description, external_references	targets, exploits

in the table I with their corresponding relationships and properties.

IV. AUTOMOTIVE NETWORKS

The shift toward intelligent and connected vehicles has led to the integration of diverse communication systems, both wired and wireless, such as Controller Area Network (CAN), Automotive Ethernet, Local Interconnect Network (LIN), Bluetooth, Cellular Communication, and Vehicle-to-X (V2X). While these communication capabilities enable real-time information exchange between vehicles, infrastructure, and pedestrians—thereby enhancing safety through early hazard detection and warning systems [18]—they simultaneously expand the attack surface, transforming contemporary vehicles into what could be characterized as Internet of Things (IoT) devices on wheels. From a cybersecurity perspective, the attack surface of modern vehicles can be systematically categorized into three primary domains based on their technological characteristics [19]:

- **Physical Interfaces:** Direct access points such as On-Board Diagnostic (OBD) ports, Universal Serial Bus (USB) connections, and Ethernet interfaces present potential entry vectors through which malicious actors may gain unauthorized access to vehicular networks.
- **Wireless Communication Channels:** The proliferation of wireless technologies including Bluetooth, Wireless Local Area Network (WLAN), and cellular communication systems exposes vehicles to remote attack vectors. These

channels can be exploited to compromise sensitive data, manipulate vehicle functions, or establish persistent unauthorized access.

- **Software Vulnerabilities:** Vulnerabilities in vehicular software systems, encompassing operating systems, application software, and firmware, constitute another significant attack domain. These vulnerabilities may be exploited to circumvent security controls, establish privilege escalation, or inject malicious code into critical vehicle systems.

A. UN R155

UN R155 is the first vehicle regulation that provides a framework for the cybersecurity of vehicles and was published in early 2021. This regulation establishes comprehensive requirements for manufacturers of member states party to the 1958 Agreement on Mutual Recognition of Vehicle Type Approval, mandating the implementation of cybersecurity considerations throughout vehicle development and approval processes. UN R155 aims to ensure robust protection against cyberattacks by requiring manufacturers to conduct systematic risk analyses, develop appropriate security measures, and demonstrate the efficacy of these measures through rigorous testing protocols. The regulation provides an extensive catalog comprising 32 common threat categories and 23 corresponding mitigation strategies within its annexes, which manufacturers must evaluate during their risk assessment procedures. Within the context of our research, these threat categories can be effectively mapped to *Attack Pattern* objects within the STIX framework, facilitating standardized representation and communication of automotive-specific threats.

B. ISO/SAE 21434: Road Vehicles - Cybersecurity Engineering

The ISO/SAE 21434 standard was developed with the primary objective of establishing a common understanding and unified methodological approach to cybersecurity engineering across the automotive supply chain. This international standard represents a collaborative effort between the International Organization for Standardization (ISO) and the Society of Automotive Engineers (SAE) to address the growing cybersecurity challenges in the automotive sector. A fundamental component of this standard is the TARA process, which provides a structured methodology for identifying and evaluating cybersecurity threats during early development phases. This proactive approach enables risk mitigation through design modifications, thereby enhancing the safety and security of vehicle occupants and other road users [3].

C. Threat Analysis and Risk Assessment

The TARA process is described in clause 15 of the ISO/SAE 21434 standard, this process involves seven steps, each with defined inputs and outputs [3]:

- **Asset Identification:** This initial phase involves the systematic identification and documentation of all assets within the system's scope that require protection. Assets encompass both physical components (ECUs, sensors,

communication buses) and intangible elements (software, data).

- **Damage Scenario:** For each identified asset, security analysts define potential adverse consequences (damage scenarios) that could result from asset compromise.
- **Threat Scenario Identification:** This step involves analyzing each asset to identify and list potential threats relevant to the automotive context. Security practitioners leverage threat catalogs such as those provided in UN R155, which delineate cybersecurity threats applicable to vehicular systems.
- **Impact Rating:** Security analysts evaluate and quantify the potential severity of consequences outlined in each damage scenario. This assessment categorizes impacts across multiple dimensions including safety, financial, operational, and privacy considerations, using a qualitative scale (severe, major, moderate, negligible).
- **Attack Path Analysis:** This phase entails the systematic analysis of potential attack progressions to realize identified threats and associated damage scenarios. This analysis often results in a visual representation like an attack tree that highlights the critical pathways an attacker might exploit.
- **Feasibility:** Security analysts assess the likelihood or practical viability of an attacker successfully executing each identified attack path.
- **Risk Assessment:** Based on calculated risk values, security practitioners determine appropriate risk treatment strategies for each threat scenario.

V. MAPPING STIX OBJECTS TO TARA

Through comprehensive analysis of cybersecurity requirements outlined in the TARA methodology, examination of STIX conceptual frameworks, and investigation of established relationships between various cybersecurity taxonomies, we have developed a systematic mapping between TARA processes and STIX objects, as can be seen in Table II.

a) **Asset Representation:** The asset identification phase of TARA involves cataloging critical components requiring protection, including ECUs, communication networks, firmware, and software systems. These assets can be represented using STIX *Infrastructure* and observable objects such as *Software*. The relationships between these assets and their organizational context can be modeled through STIX relationship types including *hosts*, *owns*, and *located-at*.

b) **Threat Modeling and Representation:** During threat scenario identification, security analysts can map identified threats to corresponding STIX *Attack Pattern* objects. This mapping leverages connections to established knowledge bases such as CAPEC, which provides structured descriptions of attack techniques, associated weaknesses, and potential countermeasures [9]. This mapping leverages connections to established knowledge bases such as CAPEC, which provides structured descriptions of attack techniques, associated weaknesses, and potential countermeasures [9]. Threat relationships can be modeled using STIX relationship types including *uses*

and *targets*, establishing connections between threat actors, attack methodologies, and targeted assets [20].

c) **Impact Assessment:** The impact rating of threats is reflected through properties within linked STIX objects, like the *Infrastructure* object and its properties such as the *infrastructure_types* or the *kill_chain_phases* which can link the infrastructure to phases of an attack. The *description* property of objects like the *Attack Pattern* and *Vulnerability* can incorporate impact assessment information, enabling security practitioners to deduce potential consequences based on standardized threat descriptions [6].

d) **Attack Path Analysis:** STIX *kill_chain_phases* property provides a structured mechanism for modeling attack progressions through different stages of cyber operations. By linking this property from STIX objects *Attack Pattern*, *Malware*, *Infrastructure*, and STIX *Tool* objects, security analysts can identify distinct attack stages and their sequential relationships. STIX relationship types employing relationship types such as *delivers*, *drops*, and *exploits* establish the directional flow of attack sequences.

e) **Feasibility Evaluation:** The assessment of attack feasibility during TARA considers multiple factors including attacker sophistication, resource requirements, and exploitation complexity. These factors can be represented through STIX object properties including *sophistication*, *resource_level*, *threat_actor_types*, *goals* properties of *Threat Actor* objects. Additional contextual information regarding attack complexity can be incorporated within the *description* property of the *Attack Pattern*, and *Vulnerability* objects.

f) **Risk Value Determination:** The risk value calculation in TARA combines feasibility and impact assessments according to methodologies defined in ISO/SAE 21434. While STIX does not directly provide risk calculation mechanisms, the values derived from previous TARA steps can be associated with corresponding STIX objects through custom properties or relationships.

g) **Risk Treatment:** Risk treatment decisions identify appropriate mitigations for identified threats based on calculated risk values. These mitigation strategies can be mapped to STIX *Course of Action* objects. STIX relationships including *mitigates* and *remediates* establish connections between mitigation measures and the corresponding threat elements represented by STIX *Threat*, *Vulnerability*, and *Attack Pattern* objects.

The diagram in Fig. 1 provides a visual representation of the relationships between the SDOs. The blue rectangles represent STIX SDOs and RSOs, and the red are the concepts coming from TARA which can be calculated and deduced from the STIX objects. The diagram showcases standard STIX relationships like *targets*, *uses*, *delivers*, *exploits*, and *mitigates*, which are used to connect STIX objects and represent interactions between them. To accommodate TARA-specific concepts that may not have direct STIX equivalents, custom relationships like *contributes-to* have been introduced. These relationships connect non-STIX objects (e.g., Impact, Risk) to STIX objects, allowing for the flow and derivation of information. In the current mapping, the security expert will

TABLE II: Mapping TARA and STIX objects.

TARA step	STIX Objects	Relationships
Asset Identification	<i>Infrastructure</i> , <i>Software</i>	<i>hosts</i> , <i>owns</i> , <i>located-at</i>
Threat Scenario Identification	<i>Attack Pattern</i> , <i>Malware</i> , <i>Vulnerability</i>	<i>uses</i> , <i>targets</i> , <i>mitigates</i>
Impact Rating	<i>Vulnerability</i> , <i>Attack Pattern</i> , <i>Infrastructure</i>	<i>contributes-to</i>
Attack Path Analysis	<i>Attack Pattern</i> , <i>Malware</i> , <i>Tools</i>	<i>delivers</i> , <i>drops</i> , <i>exploits</i>
Attack Feasibility Rating	<i>Threat Actor</i> , <i>Attack Pattern</i> , <i>Vulnerability</i>	<i>contributes-to</i>
Risk Value Determination	Feasibility * Impact	<i>contributes-to</i>
Risk Treatment Decision	<i>Course of Action</i>	<i>mitigates</i>

be required to use the STIX object's properties to assess the impact and feasibility by considering properties like *confidence*, *threat_actor_types*, *sophistication*, *resource_level*, *primary_motivation*, *external_references* and *kill_chain_phases*.

VI. DISCUSSION

The integration of STIX with TARA methodology offers significant potential for enhancing automotive cybersecurity practices through standardized threat intelligence representation. Key benefits include:

- Enhanced threat modeling: Mapping between TARA threat scenarios and STIX Attack Pattern objects establishes formal connections between automotive-specific threats and standardized threat intelligence, providing security practitioners with improved visibility into adversary TTPs. For example, a TARA threat scenario describing "Spoofing of a signal leads to loss of integrity of the data communication" can be mapped to specific STIX Attack Pattern objects that delineate signal spoofing techniques, their implementation vectors, and associated indicators.
- Structured mitigation strategies: The correlation between TARA risk treatment decisions and STIX Course of Action objects enables precise representation of security controls, linking requirements like message authentication to specific cryptographic mechanisms and protocols.
- Standardized communication: STIX establishes a shared vocabulary and data model across organizational boundaries while enabling automated information sharing and integration with security tools, threat intelligence platforms, and SIEM solutions.
- Regulatory compliance support: The mapping facilitates documentation demonstrating that security assessments have considered relevant attack patterns and vulnerabilities, supporting compliance with regulations such as UN R155 [4].

Despite these potential benefits, several challenges must be addressed for effective implementation. STIX was originally designed for IT environments and may require adaptation to fully address the specific context of automotive systems [21].

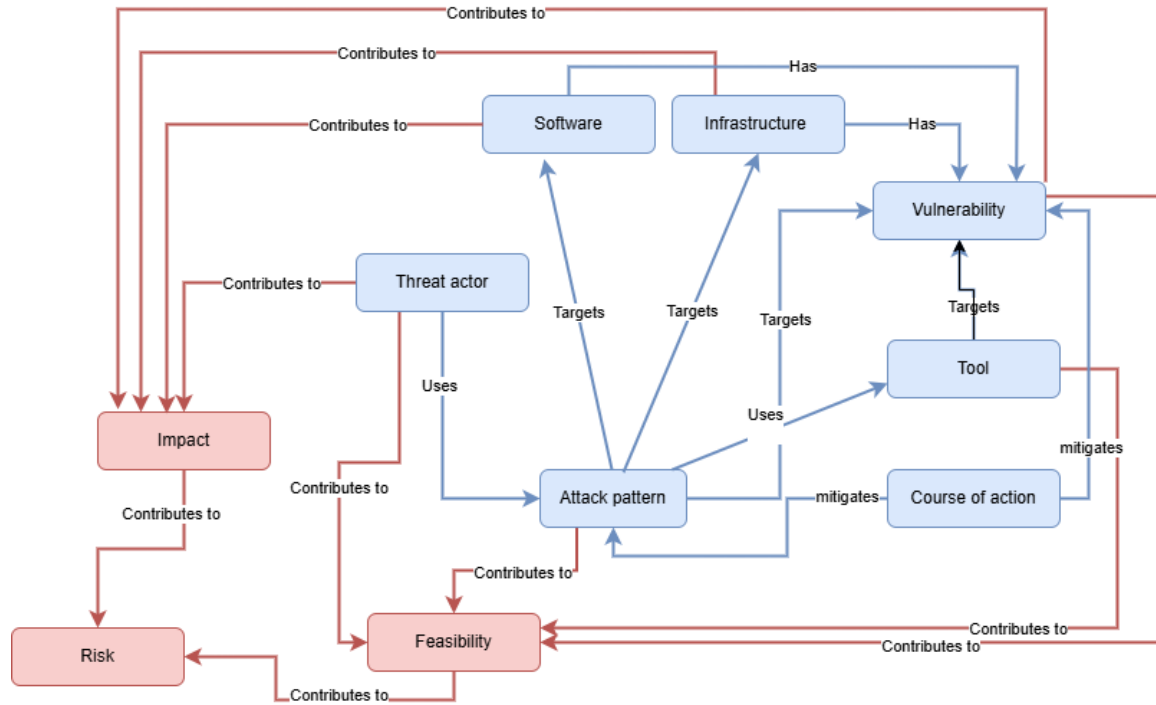


Fig. 1: Visualization of the relationships between the SDOs and TARA objects.

The automotive cybersecurity domain lacks standardized terminology, potentially causing inconsistencies when mapping TARA objects to STIX constructs [22]. Additionally, integration complexity may require specialized expertise, and maintaining alignment between long automotive development cycles and rapidly evolving threat intelligence presents ongoing challenges [23]. Nevertheless, the potential benefits justify continued research into standardizing automotive cybersecurity intelligence through frameworks like STIX, advancing toward more comprehensive, efficient security practices across the industry.

VII. CONCLUSION

This research has examined the mapping between the STIX framework and the TARA methodology mandated by ISO/SAE 21434 for automotive cybersecurity engineering. Our analysis demonstrates that this integration offers substantial benefits for enhancing automotive cybersecurity practices across complex supply chains and diverse stakeholder ecosystems. The proposed mapping enables automotive security practitioners to develop more precise characterizations of threats and vulnerabilities while providing detailed technical information for focused risk assessment and mitigation planning. Significantly, this integration facilitates seamless exchange of cyber threat intelligence among diverse stakeholders in the automotive ecosystem through a standardized representation of threats, vulnerabilities, and mitigations. Despite the identified benefits, implementation challenges include the need for automotive-specific extensions to existing STIX objects to capture the unique characteristics of vehicular systems. In

future research, we intend to develop and evaluate ontologies specifically designed for the automotive cybersecurity domain. These ontologies will provide formal representations of domain-specific concepts, entities, relationships, and attributes, enabling more precise and unambiguous sharing of CTI across the automotive ecosystem.

VIII. ACKNOWLEDGMENT

The authors would furthermore like to thank the AIAM (Applied Informatics and Applied Mathematics) doctoral school of Obuda University, Budapest, Hungary for their support in this research. The research was Supported by the 2024-2.1.1 University Research Scholarship Program of the Ministry for Culture and Innovation from the Source of the National Research, Development and Innovation Fund. We would like to thank the Robert Bosch Ltd./Bosch Group, Hungary, for their professional support of the research.

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